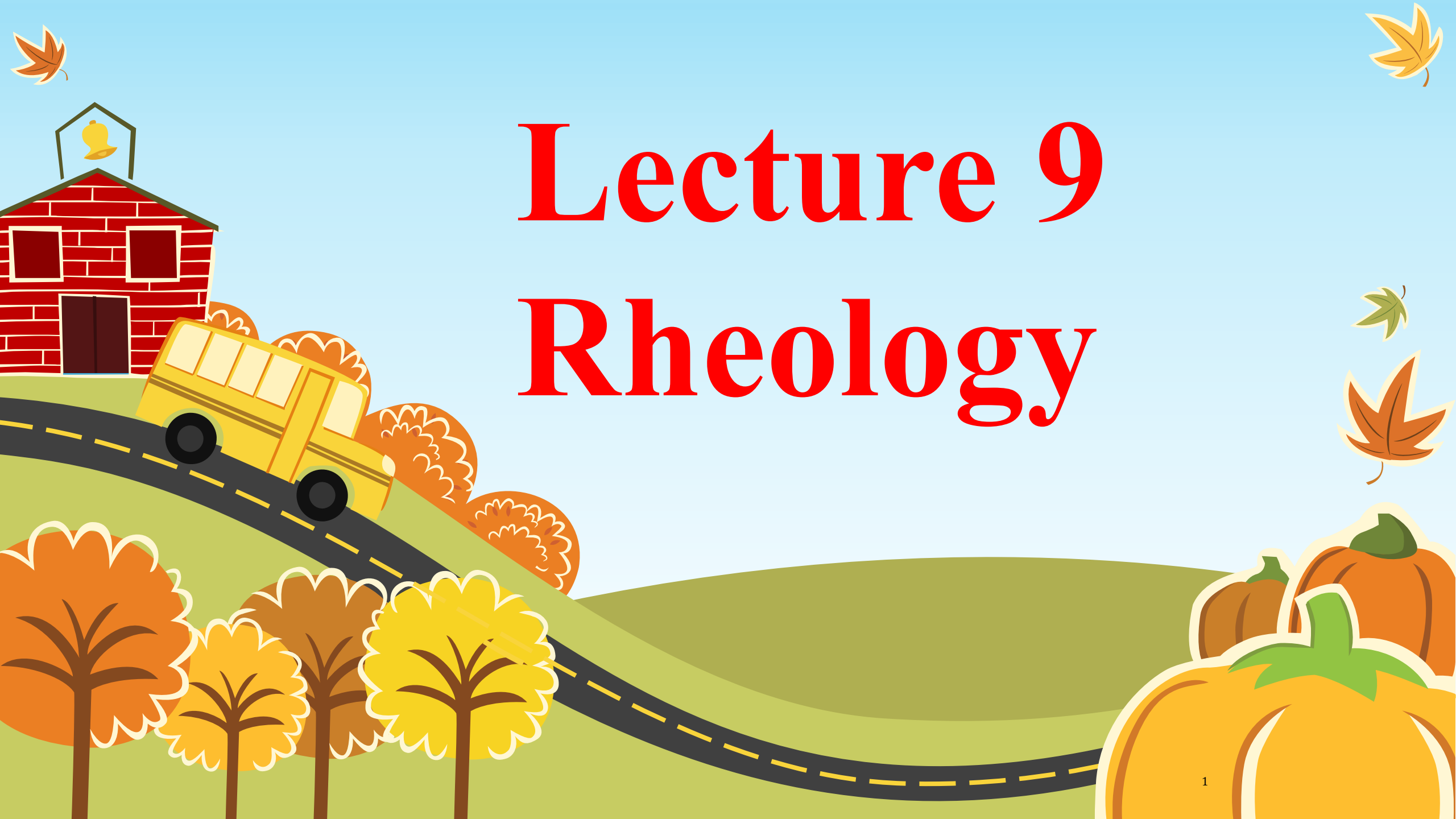


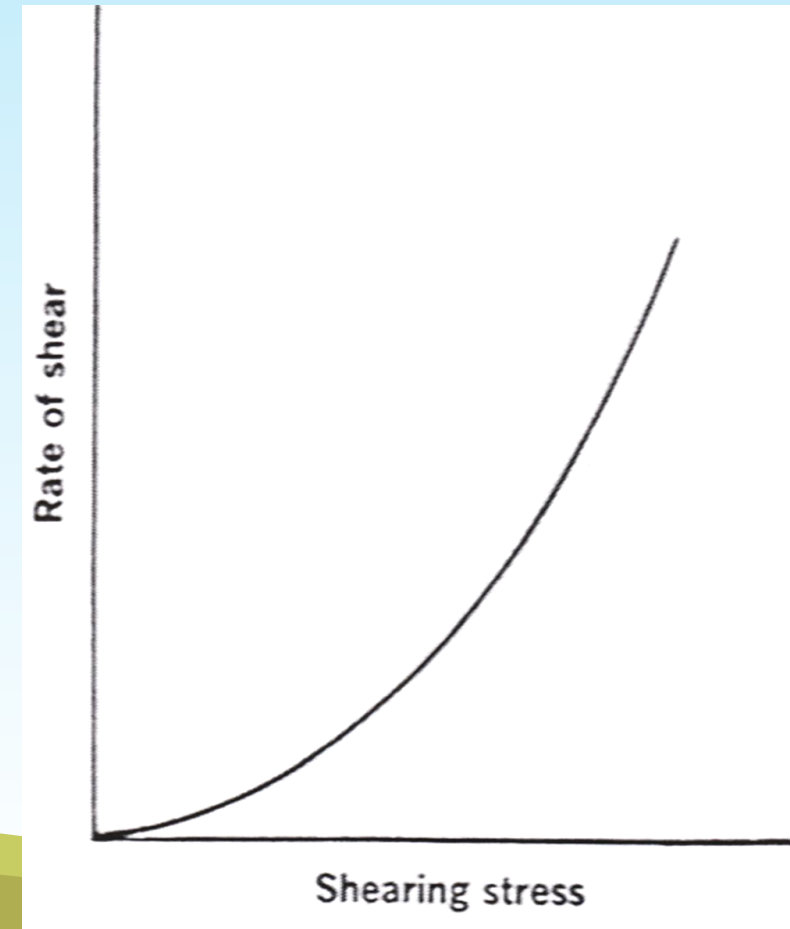
Lecture 9

Rheology



2- Pseudoplastic Flow **Shear Thinning**

- At start the material is characterized by high viscosity latter upon shear the **viscosity decreases (Shear Thinning)**
- The curve for a pseudoplastic material begins at the origin 📢
- Ex: **linear polymers**. (gums as tragacanth & sodium aliginate & carboxymethyl cellulose)
- **Viscosity at any point is determined from the tangent line of the curve at that point**





Long chain
polymers coiled
and entrapping an
amount of the
vehicle within the
coils



Shear applied



Alignment of the
polymer chains and
release of the entrapped
vehicle resulting in
easier flow (higher rate
of shear in response to
applied stress) and
reduction in the internal
resistance

Example: ketchup bottle



Pseudoplastic systems which cannot be described by a single value for viscosity are expressed by:

$$F^N = \eta' G$$

When $N = 1$, equation the flow is Newtonian .

When N has a value not equal to 1, the flow becomes increasingly non Newtonian.

if $N > 1$ pseudoplastic

The term η' is a viscosity coefficient.

The logarithmic form is a straight line equation

$$\log G = N \log F - \log \eta'$$

A straight line is obtained when $\log G$ is plotted against $\log F$



Thixotropy

Relieving stress

stress



Gels or thick
fluids

Require a fixed time to
return

Thin, less viscous

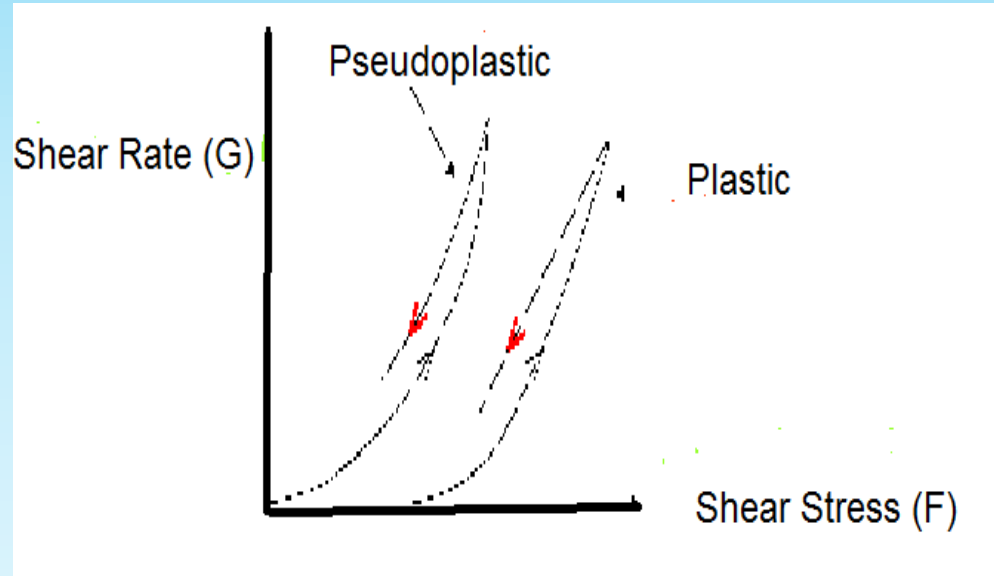
Gel → Sol → Gel

Rheogram

A thixotropic fluid is a fluid which takes a finite time to attain equilibrium viscosity

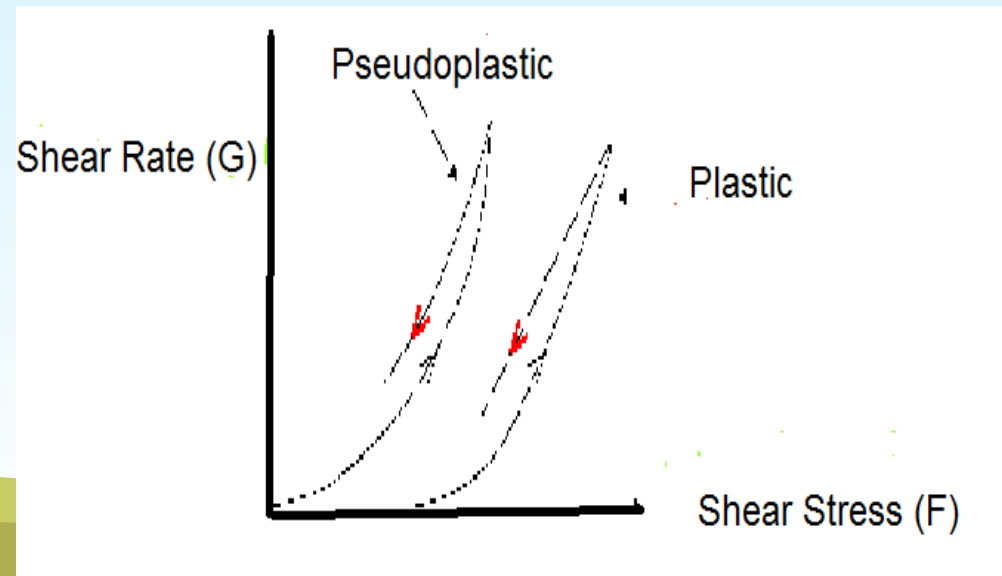
→ Some thixotropic fluids return to a gel state almost instantly, and are called pseudoplastic fluids.

→ *all materials that are shear thinning are thixotropic, in that they will always take a finite time to bring about the rearrangements needed in the microstructural elements that result in shear thinning.*



Measurement of Thixotropy

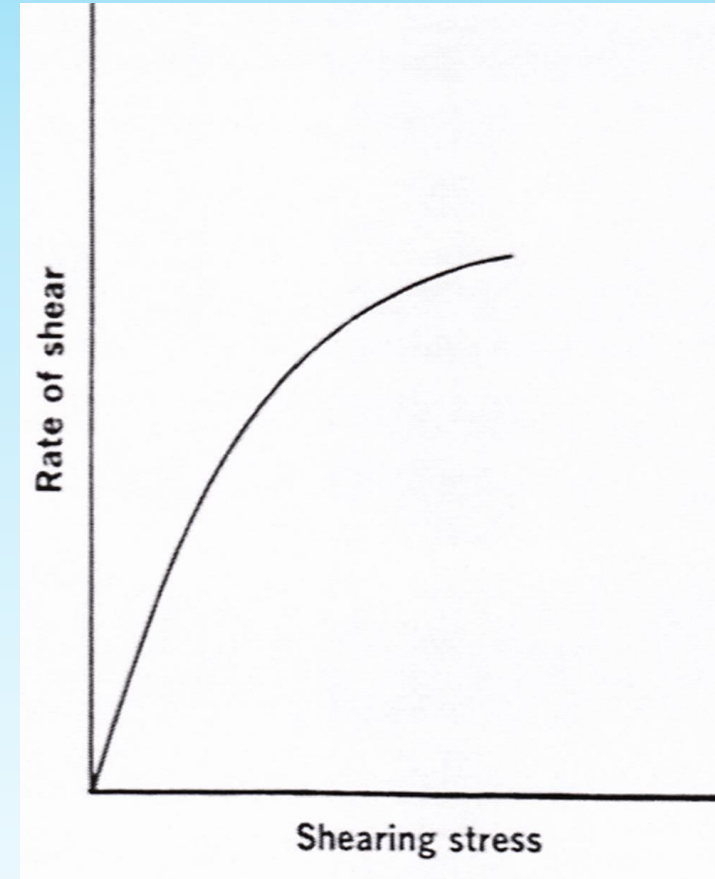
- By measuring the **hysteresis loop** (area between the up- and down-curves of the rheogram).
- it is a measure of thixotropic breakdown



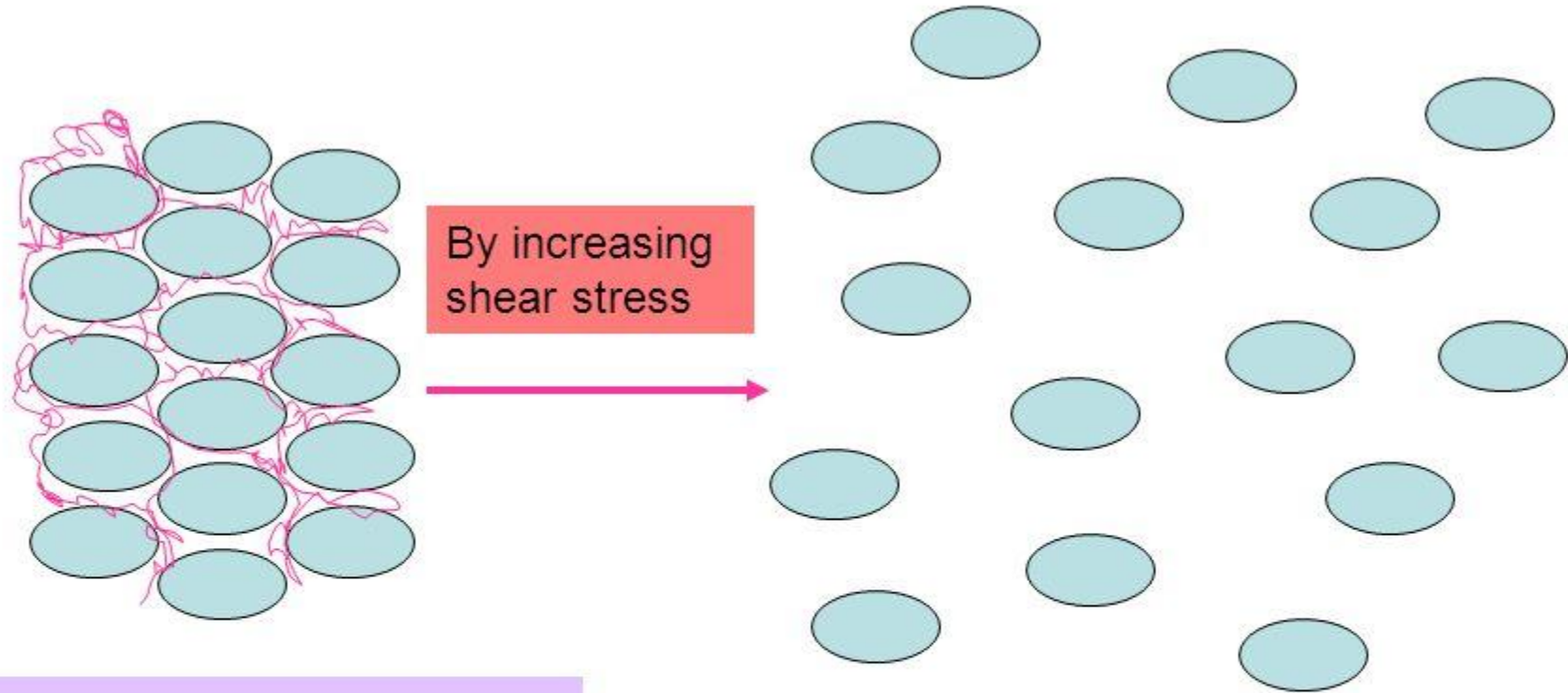
- **Example** aqueous dispersion of 8% w/w sodium bentonite (flocculated suspension)
- In the beginning it is in the **gel** form (high viscosity) after stirring above the yield value the viscosity decrease can be poured (**sol**) after removal of shear it reform a **gel** again
- Gel  Sol  Gel transformation

3- Dilatants Flow

- The curve for a dilatant material begins at the origin
- At start the material is characterized by lower viscosity latter upon shear the **viscosity increases (Shear Thickening)**
- Such systems actually increase in **volume** when sheared & are called dilatants.
- Ex: **deflocculated suspensions** with a high percentage of dispersed solids exhibit an increase in resistance to flow with increasing rates of shear.
- When the stress is removed, a dilatants system returns to its original state of fluidity.
- $F^N = \eta' G \quad (N < 1)$



The dilatant flow can be explained as follows



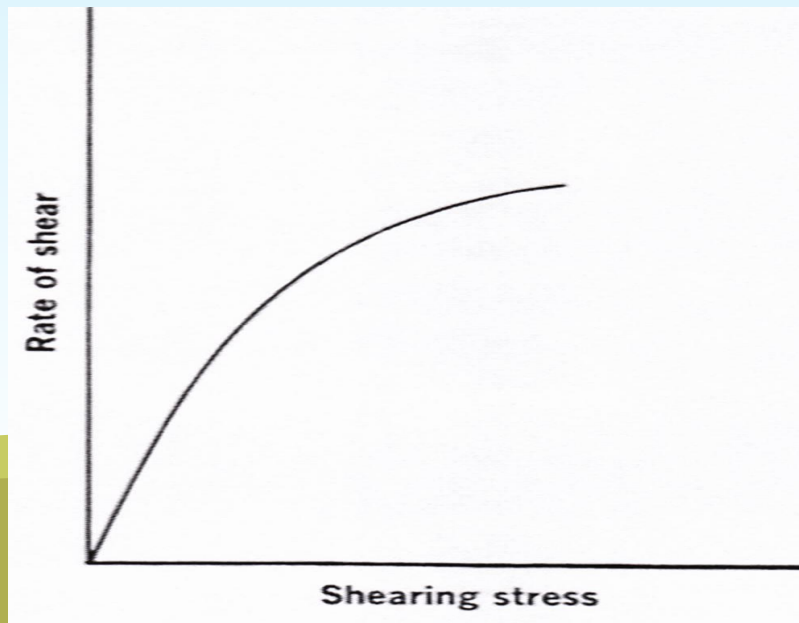
*Closed – packed particles,
minimum void volume,
sufficient vehicle,
relative low consistency*

Open packed (dilated) particles,
Increased void volume,
Relative high consistency

- *at rest* → particles → packed with **minimum void volume** and volume of **vehicle sufficient to fill** these voids facilitating particle motion of the material (low viscosity)
- *at high shear* → particles move quickly over each other taking an **open packing** with **large voids between the particles** (hence the term dilatant , increase in volume)
- the amount of vehicle which **wets and lubricate** the particles now is **insufficient** to fill all these voids so there is increasing the resistance of flow of particles over each other (increase in viscosity)
- so take care upon using a mixer or blender as this material may solidify and damage the mixer

Shear thickening behavior is highly dependent upon **the volume fraction of solid particulate suspended within the liquid**. The higher the volume fraction, the less shear required to initiate the shear thickening behavior.

i.e: increasing the solid concentration results in an increase in the degree of dilatancy



$$\tau = \eta' \dot{\gamma}^N$$

$N < 1$ and decreases as the degree of dilatancy increases.

As N approaches 1, the system becomes increasingly Newtonian in behavior.

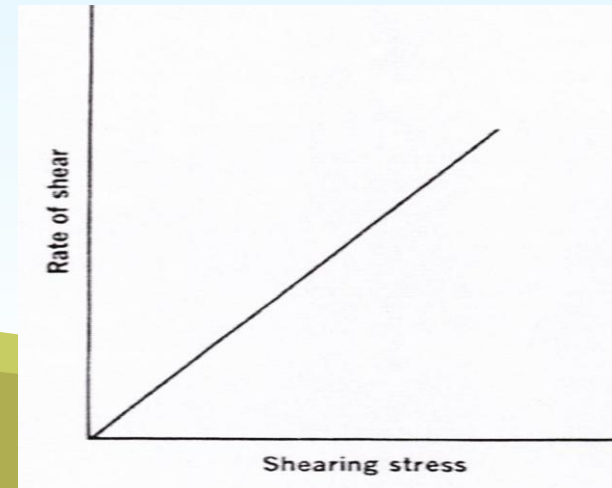
Instrumentation to measure viscosity

Viscometer	
Single/One point: Measures the viscosity at a single rate of shear → one point on the curve	Multipoint: Measures the viscosity at several rates of shear → many points on the curve
Equipment: 1) Ostwald viscometer 2) Falling sphere viscometer	Equipment: 1) Cup and bob 2) Cone and plate
Applications: • Newtonian fluids only	Applications: • Non-Newtonian fluids • Newtonian fluids

Instrumentation

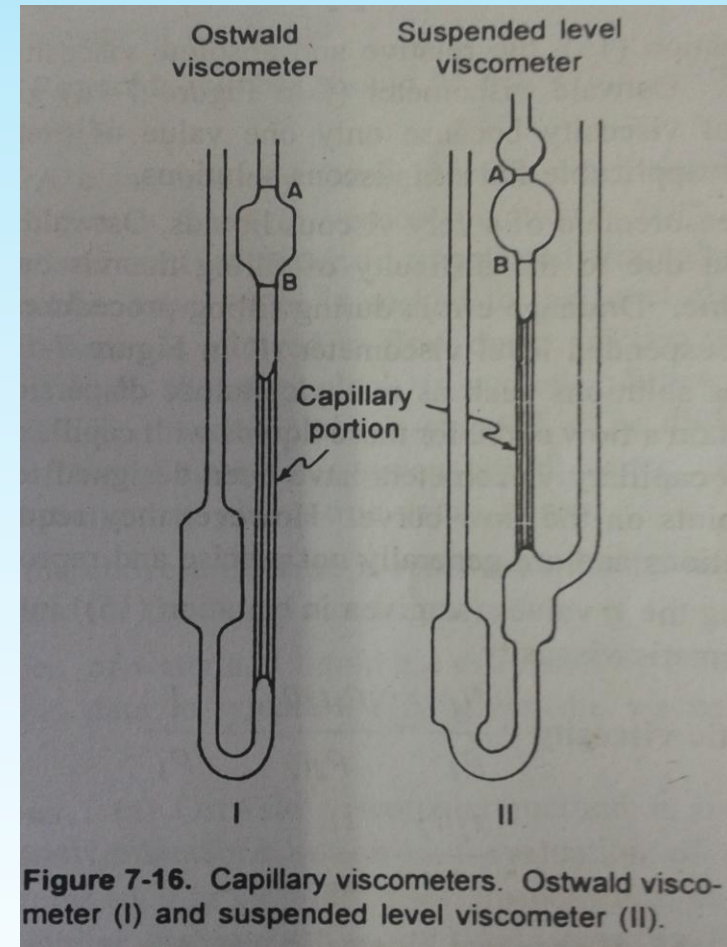
“One point” instruments

- Provide a single point on the rheogram.
- Extrapolation of a line through this point to the origin will result in the complete rheogram.
- The capillary and falling sphere are for use only with **Newtonian materials**.

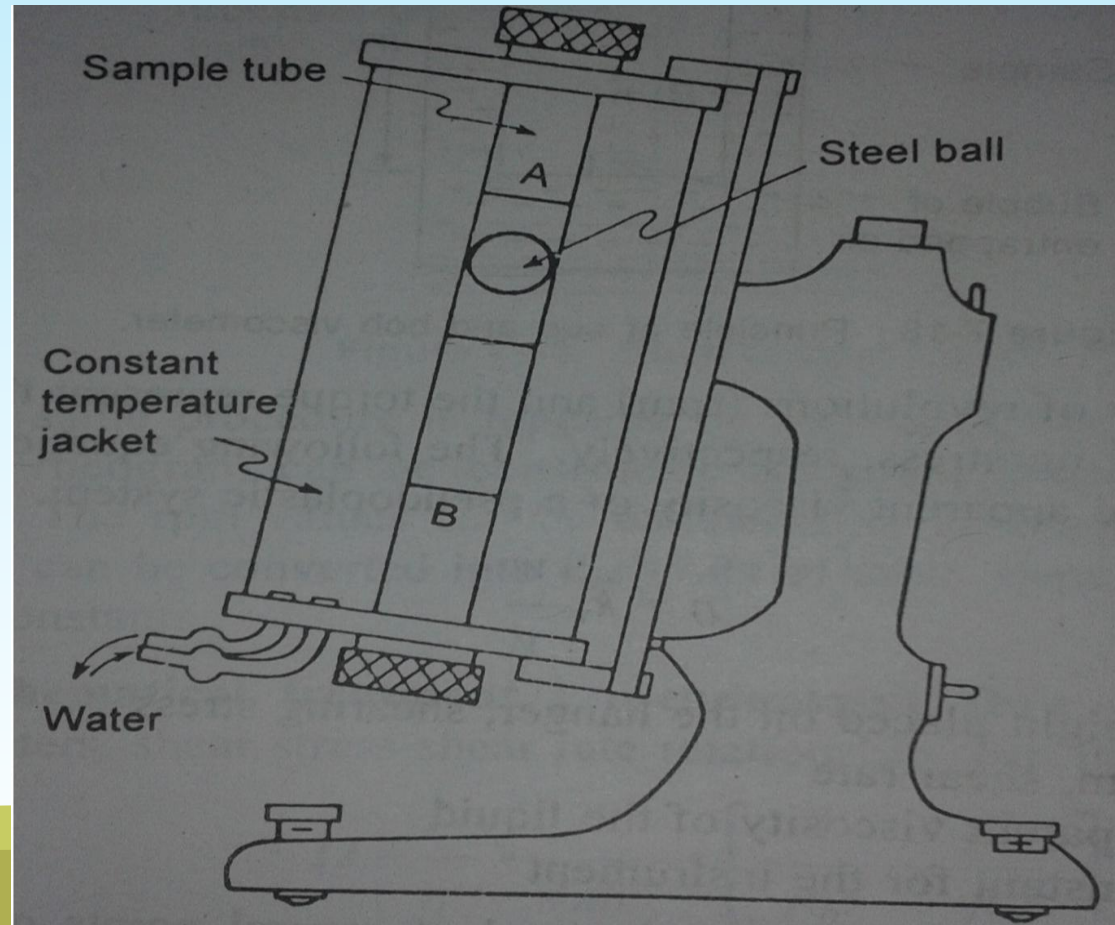


Ostwald Viscometer

- Ostwald viscometer is used to determine the viscosity of a Newtonian liquid.
- **The higher the viscosity of a liquid the more the time it takes to pass through a capillary tube**
- When a liquid flows by gravity, the time required for the liquid to pass between two marks (A and B shown in Figure) through a vertical capillary tube is determined.

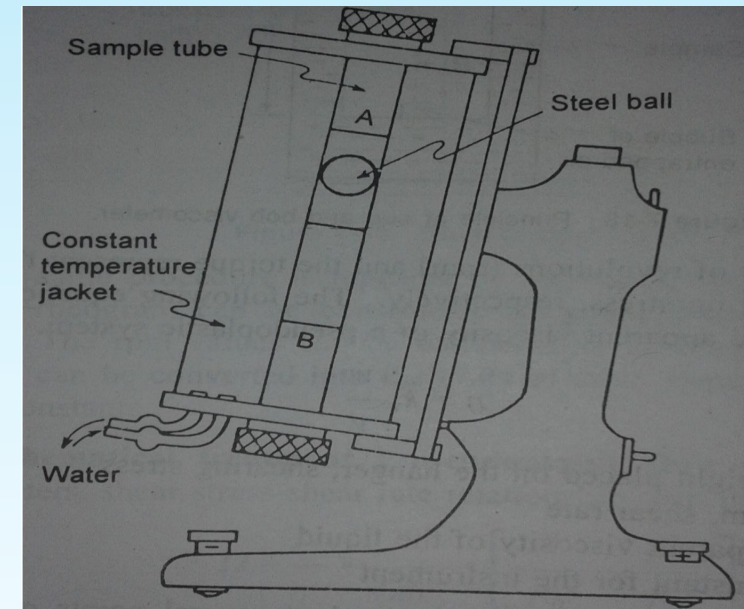


Falling Sphere Viscometer



Falling Sphere Viscometer

- The sample & ball are placed in the inner glass tube & allowed to reach temperature equilibrium with the water in the surrounding constant temperature jacket.
- The tube & jacket are then inverted, which effectively places the ball at the top of the inner glass tube.
- The time for the ball to fall between two marks is accurately measured & repeated several times.
- The higher the viscosity the longer it would take for the ball to travel through the liquid



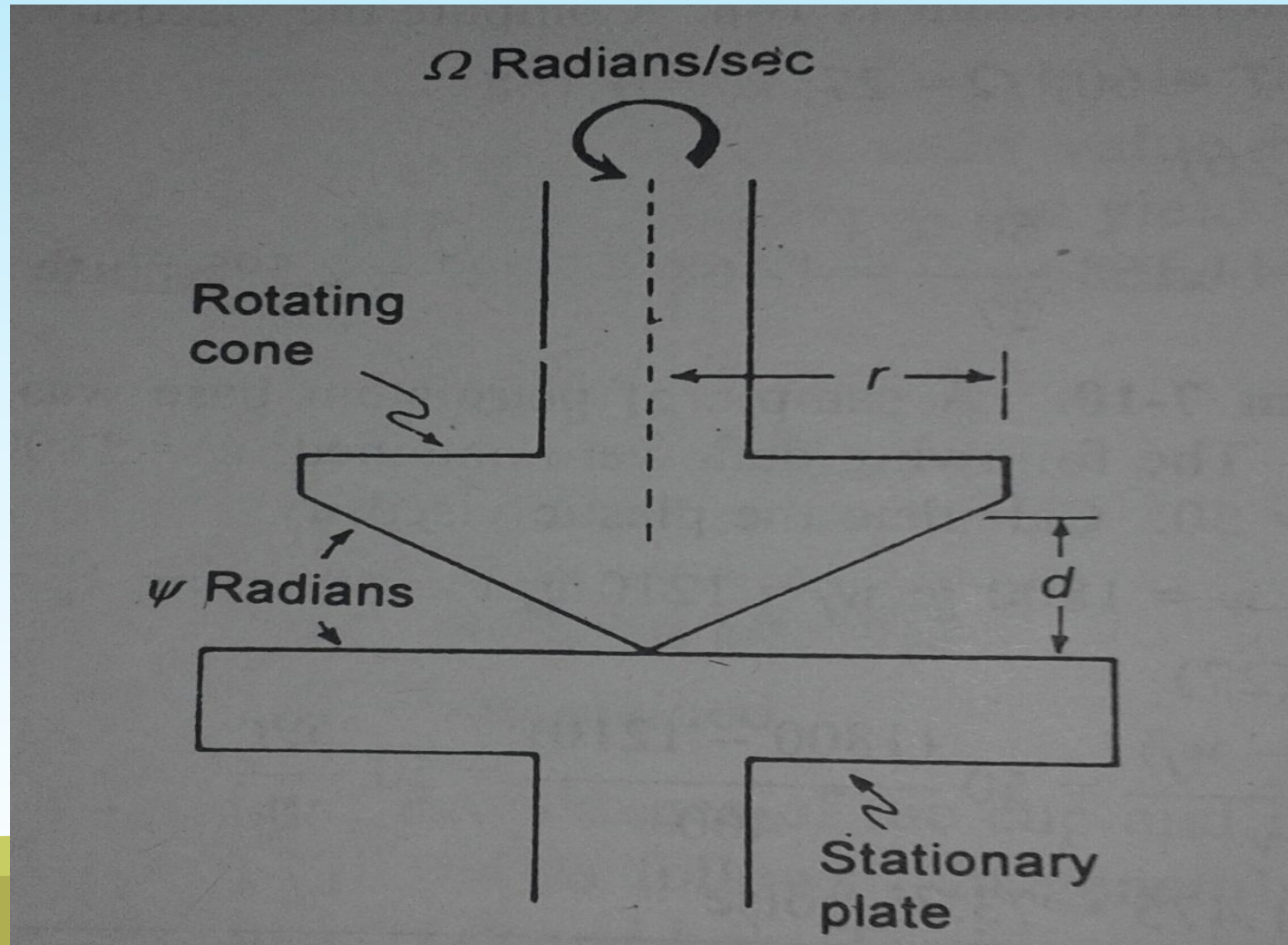


Instrumentation

“Multi-point” instruments

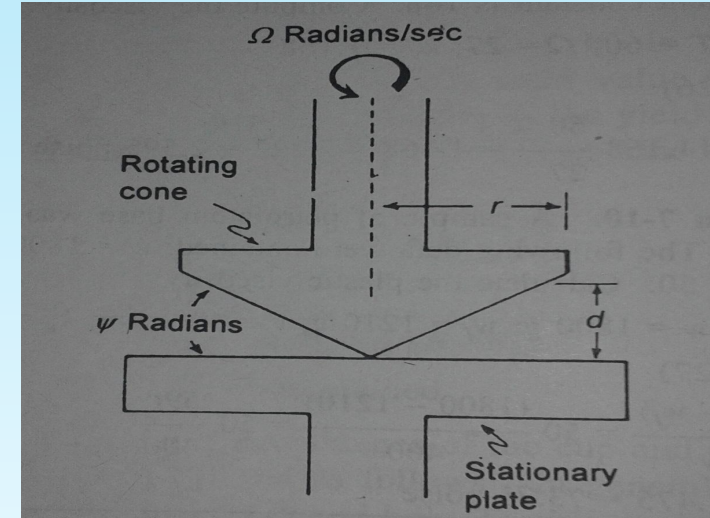
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- 
- Used with non-Newtonian systems.
 - The instrumentation used must be able to operate at a variety of rates of shear.
 - Cup and Bob , Cone and Plate viscometers may be used with both types of flow system.

Cone and Plate Viscometer



Cone and Plate Viscometer

- The sample is placed at the center of the plate which is then raised into position under the cone.
- The cone is driven by a variable speed motor & the sample is sheared in the narrow gap between the stationary plate and the rotating cone.
- The **rate of shear** in rev. /min. is increased & decreased by a selector dial & the **torque** (shearing stress) produced on the cone is read on the indicator scale.
- A plot of rpm or rate of shear versus scale reading (shearing stress) may be plotted.





Factors Affecting Rheological Properties in Pharmaceutical Products



- **1 - Chemical Factors**
 - **(a) Degree of Polymerization**
 - The **higher the molecular weight the higher will be the viscosity**
 - So high molecular weight compounds are used in small concentration to get the desired viscosity
 - **(b) Extent of Polymer Hydration**
 - Hydrophilic polymer are associated with water molecules (immobilized) forming solvent layer
 - this hydration increase the viscosity.
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- **(c) Impurities, Trace Ions and Electrolytes**

- Impurities can change the viscosity of natural polymers,

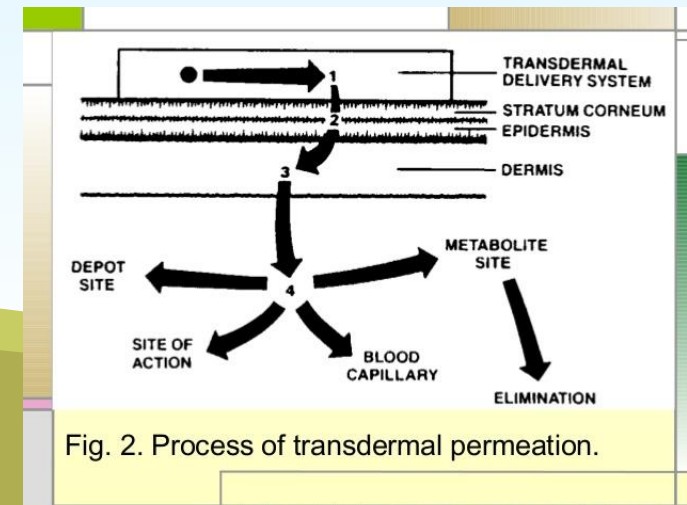
- e.g. Sodium Alginate (sol) + Calcium  Calcium Aliginate (Gel)

- 
- 
- **Electrolytes** at low concentration they do not change the viscosity
 - At high concentrations, they **compete for adsorbed water (around the hydrated polymer) leading to polymer precipitation and decrease in viscosity**
- 
- 

Pharmaceutical Applications

- **(1) Prolongation of Drug Action**
- **40 – 70 % Procaine Penicillin G + sod citrate + polysorbate 80 → IM
→ thixotropic & depot (prolonged effect in blood)**
 - **Gel → Sol → Gel**
- **the viscosity decrease upon shearing (injecting the suspension) →
after injecting into the muscle → it form a cohesive spherical depot
(because of increase of its viscosity due to decrease the shear)**
- **These depots resist disintegration and absorption of penicillin (high
viscosity and small surface area) so it have a prolonged blood level (4
days)**
- **This cohesive depot needs a high yield value & rapid thixotropic
recovery**

- **(2) Effect on Drug Absorption**
- In creams and lotions the viscosity affect the rate of absorption in the skin
- the lower the viscosity the more the amount released or diffused from the cream to the skin



(3) Thixotropy and Suspension Formulation

- In suspension or emulsion it is desirable to formulate it in high viscosity to decrease the sedimentation rate of suspension or the creaming rate of emulsion (increase stability)
- But this high viscosity would make it hard to pour out of the container
- Upon using a thixotropic material → example sodium bentonite ,colloidal silicon dioxide we can dissolve the problem
- **On rest → high viscosity (gel) higher stability**
- **on shear (shaking the bottle) → (sol) low viscosity easy pouring**
- **on rest (on the shelf) → (gel) increase in viscosity and no caking or creaming**